AI ACADEMY · LEVEL 2 · GRADE 7 · WEEK 16 · STUDENT LESSON

Overfitting Intuition

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

Before you start

Week 11: judge transfer on cases not used to choose the rule.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Compare training and validation evidence for model capacity. Try the recall before opening notes.

Your learning target

I can compare **training** and validation **evidence** for model **capacity**.

I can explain **overfitting** without assuming every gap proves it.

I can use a declared selection rule instead of perfect **training** fit.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Choose Informative Questions. Give an example and a mistake that the idea helps you avoid.

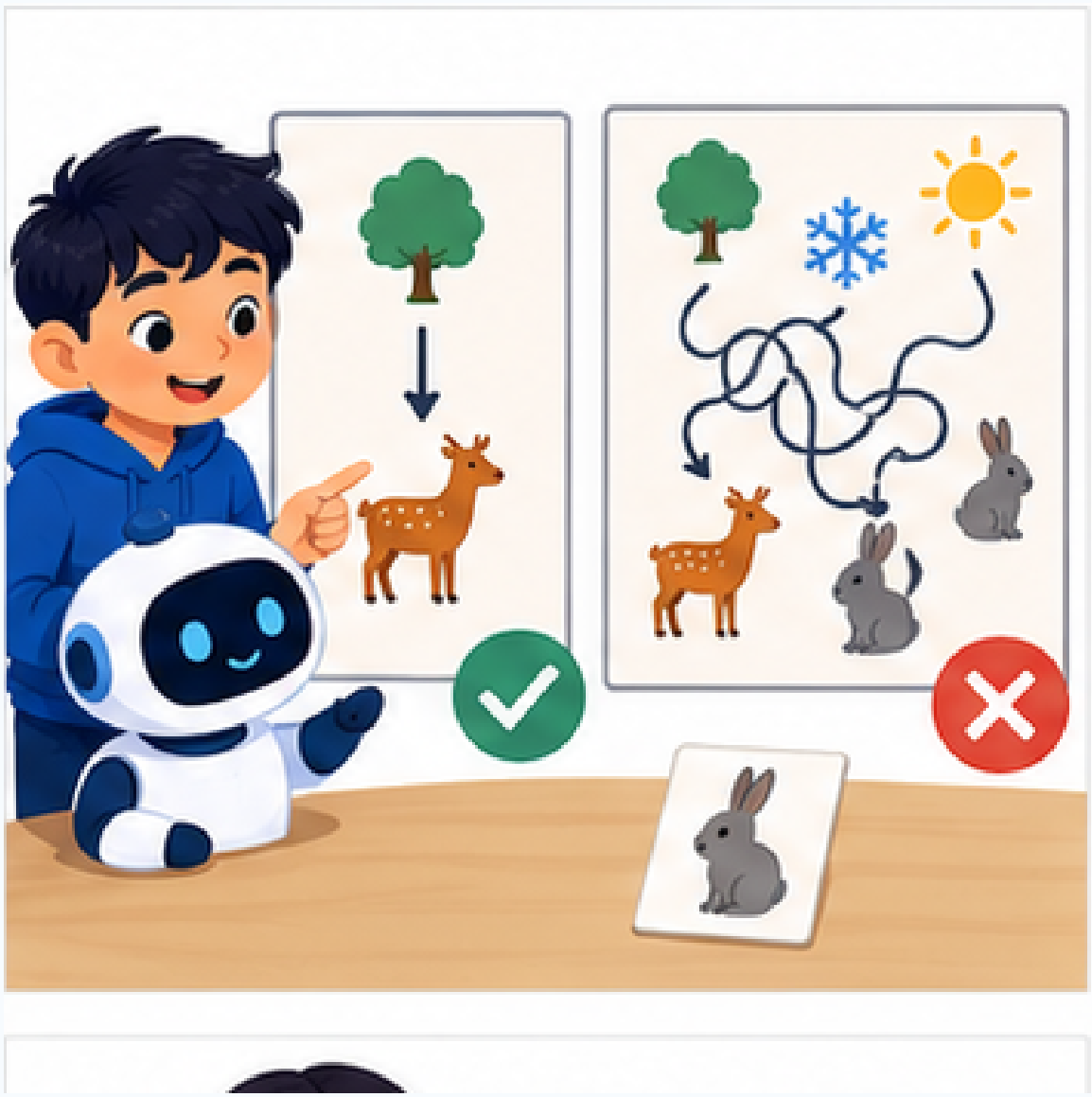
Without opening the old lesson, explain one key idea from Types of Classification Error. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Human Classifier Simulation. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 16: **Overfitting** Intuition

A rule that memorizes examples may fail on new cases.



THINKER MISSION

Contrast a simple rule with a seven-**exception** rule.

Thinker name and date:

Write or draw your reasoning and evidence.

Readiness for the connected explanation

Can a zero **training** error guarantee good validation performance?

This week's end goal

By the end, I can use development performance rather than **training** fit alone to choose model **complexity**.

STUDENT LESSON · 2 OF 18

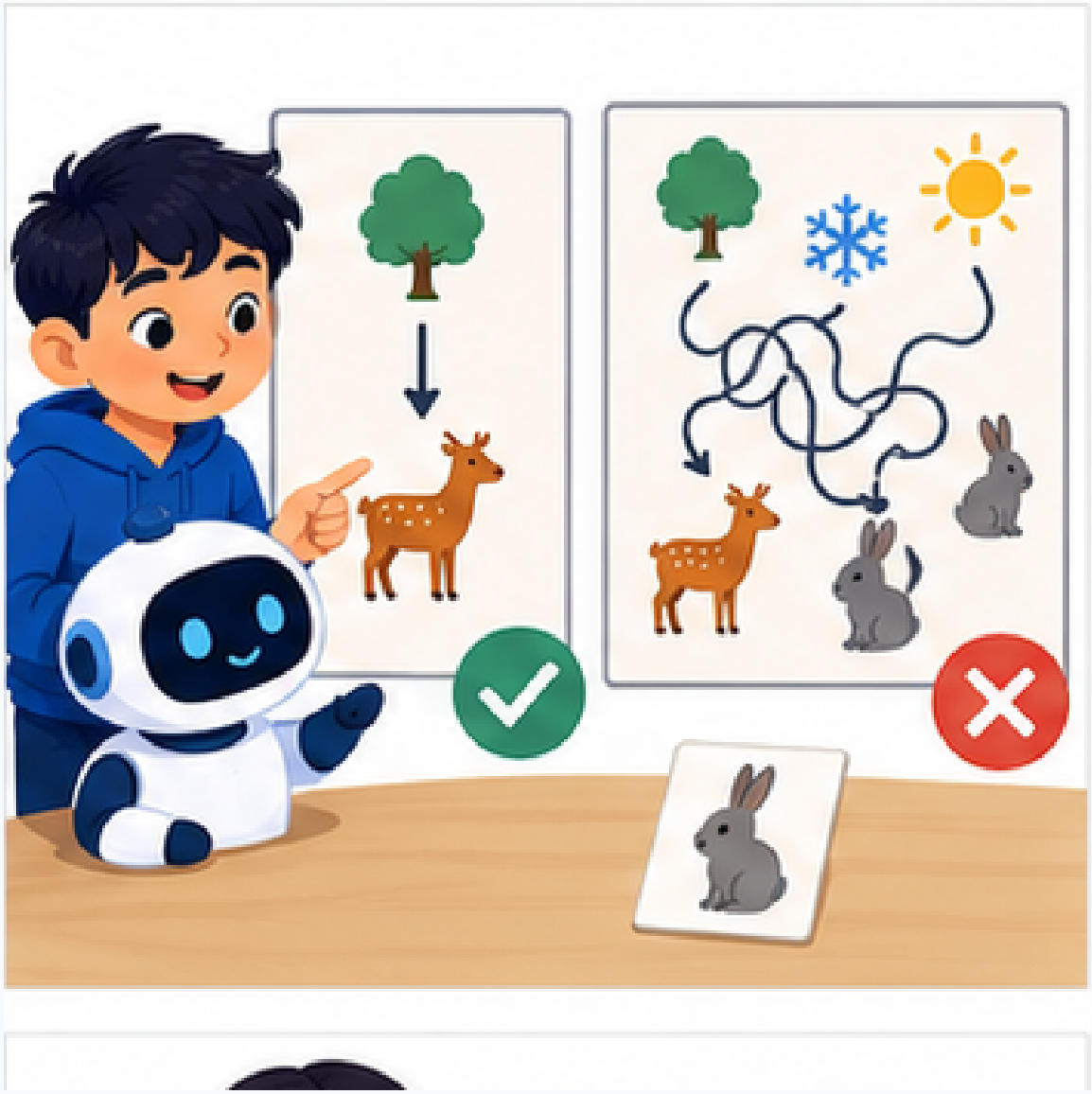
Notice the evidence

Investigate the supplied Overfitting Intuition case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



Start with this puzzle

MYSTERY

The learners must use **overfitting** intuition to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

Write or draw your reasoning and evidence.

My **prediction** and the clue supporting it:

Write or draw your reasoning and evidence.

A closer explanation

A more flexible model can represent complicated relationships and accidental details. **Capacity** can help when there is enough relevant **evidence**, but perfect **training** performance is not the goal.

Compare **training** and validation using the same metric while checking leakage and representativeness.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason.

Earlier weeks supplied the model-building **process**; this week examines **overfitting** and model **capacity**.

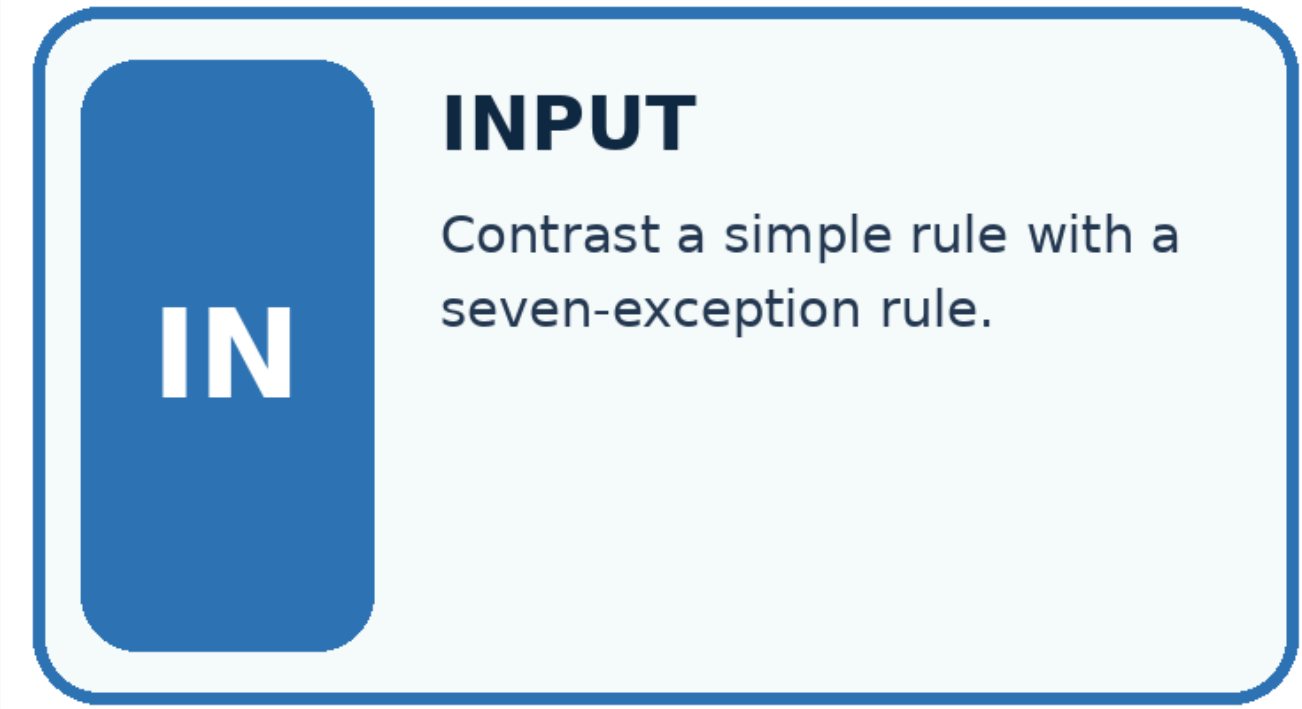
See how it works

> Follow the mechanism for Overfitting Intuition. Connect each labelled input, deciding step and result.

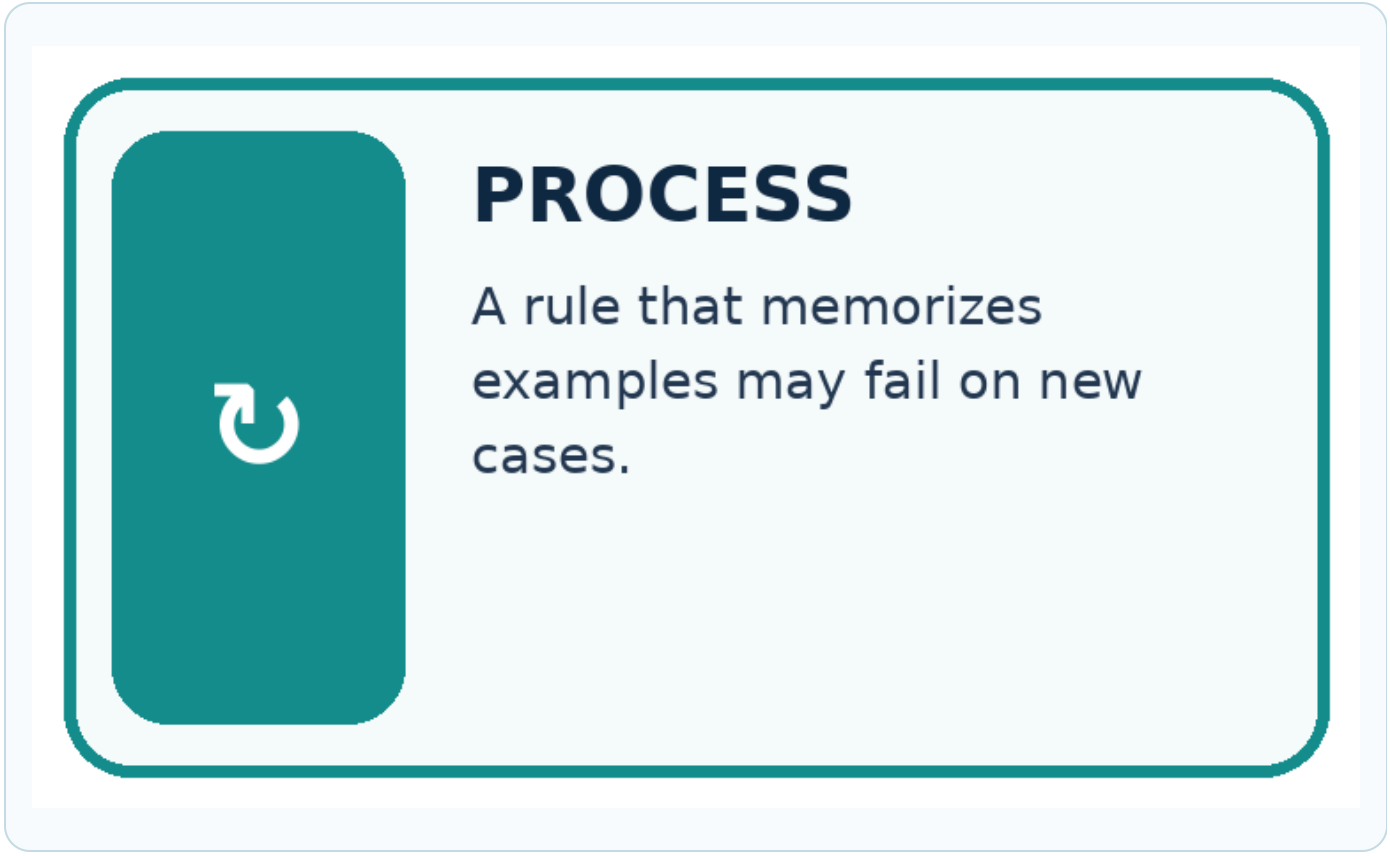
VISUAL MECHANISM

Input → Process → Output

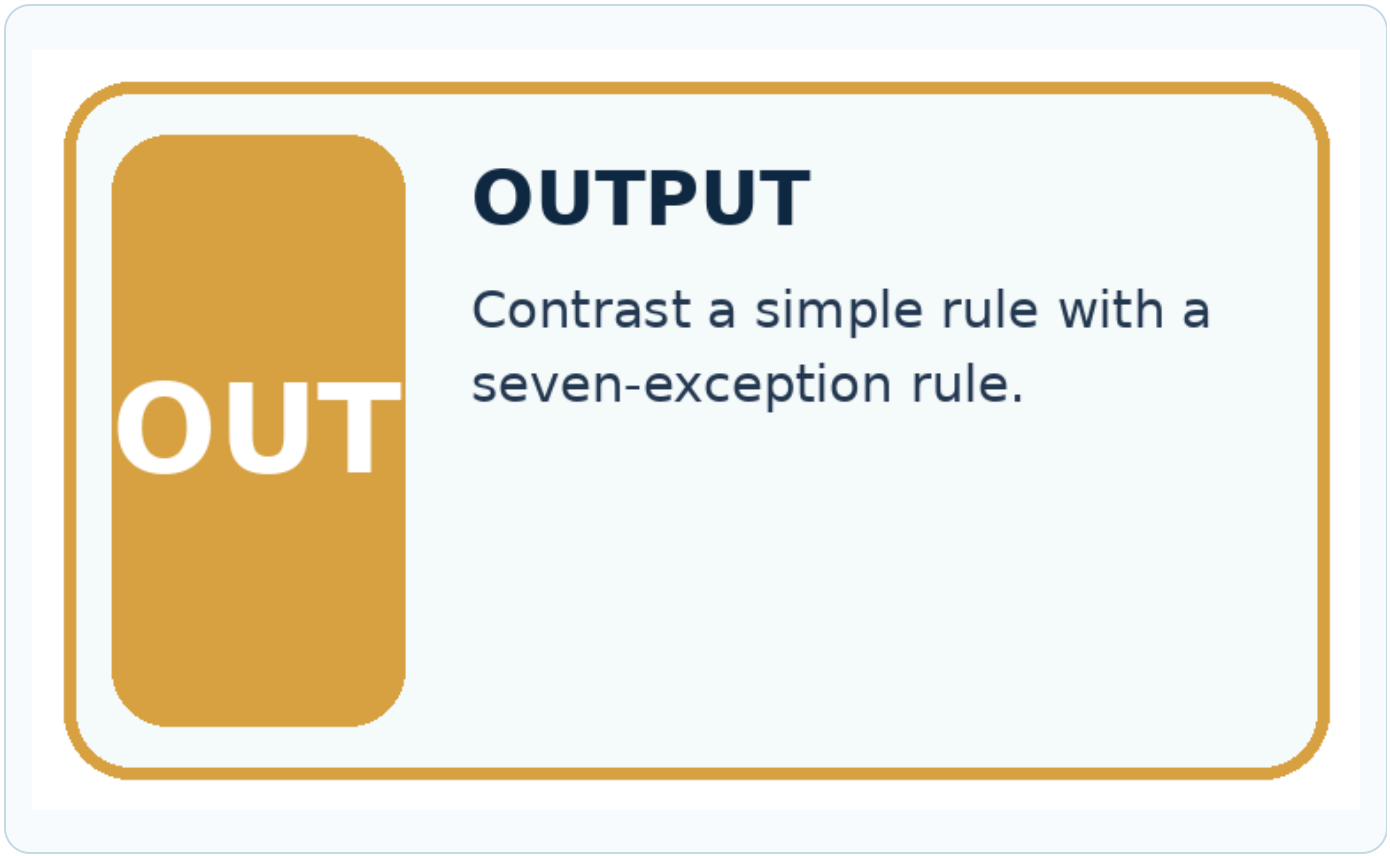
Trace the information. At each stage, ask what changed and how you could test it.



↓


PROCESS
A rule that memorizes examples may fail on new cases.

↓


OUT
OUTPUT
Contrast a simple rule with a seven-exception rule.

Which stage could fail even when the other two are correct? Explain:

> **Write or draw your reasoning and evidence.**

A closer explanation

MAE means mean absolute error: find the size of each prediction mistake without its plus or minus sign, then average those sizes. Lower MAE means smaller average mistakes in the stated units. On the same regression split, a constant has **training** MAE 5 and validation 6, a line has **training** 2 and validation 3, and a highly flexible curve has **training** 0 and validation 8. Under smallest validation MAE, choose the line.

The curve has the lowest **training** loss but the worst validation result.

Before continuing, underline the decision rule and circle the **evidence** that supports the conclusion. Explain the example to a partner without replacing the counts or conditions with a stronger claim.

Keep the distinction between an observed result and a proposed next step.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

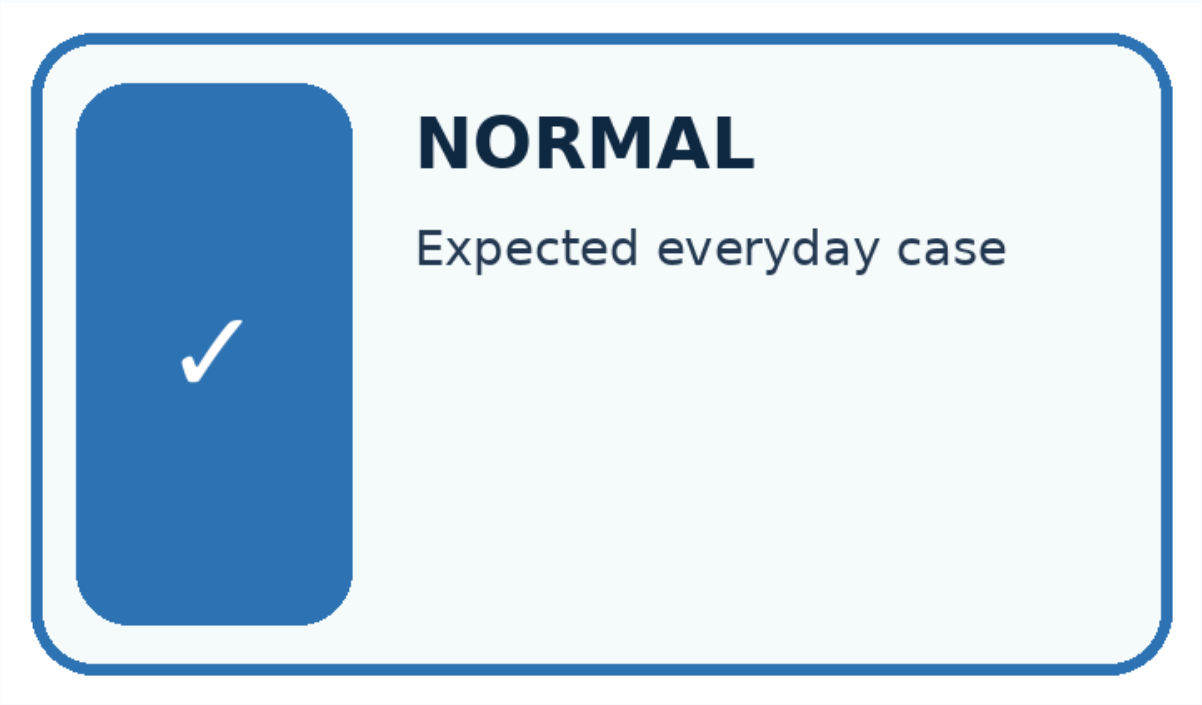
> Compare the supplied cases for Overfitting Intuition. Identify the detail that changes the conclusion.

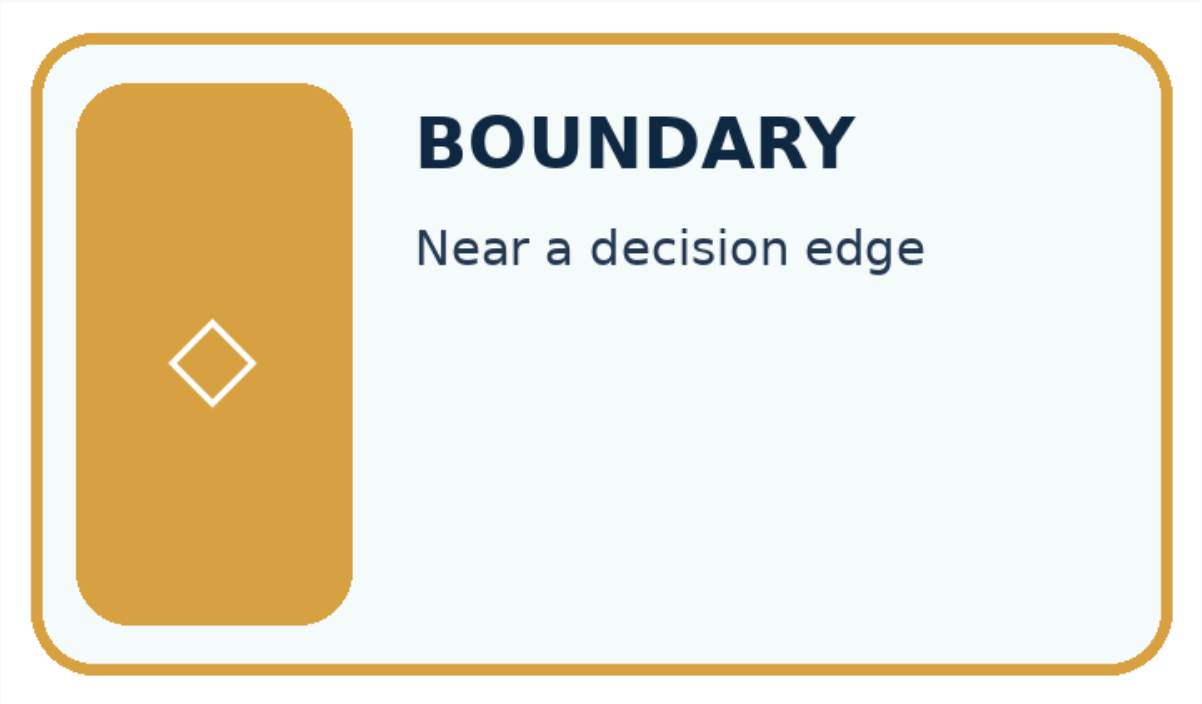
The case to investigate

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

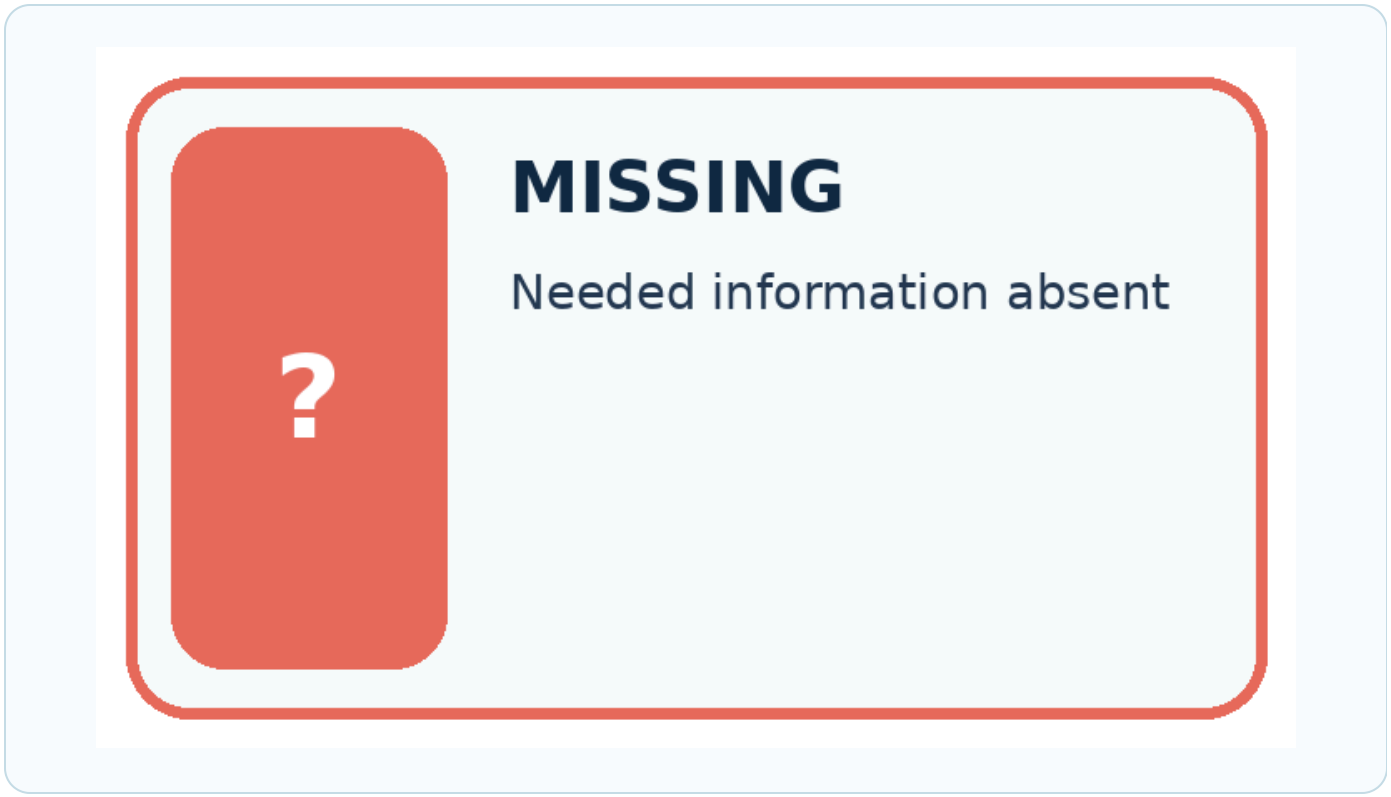
A system is not understood until it survives more than one comfortable example.





The normal case

NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.


?
MISSING
Needed information absent


→
TRANSFER
Fresh situation or user

Check missing information

MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change one thing

Change: Complex's validation error becomes 2 on a newly documented, fair development comparison.

Predicted result

Prediction: The stated rule now selects Complex.

Why the result follows

Reason: The rule compares validation errors; it does not ban complex models.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

> Use the vocabulary of Overfitting Intuition to explain a concrete decision, rather than only memorising definitions.

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.




2
GENERALIZE
work on new relevant cases
3
COMPLEXITY
number of conditions or exceptions



Word	My own example	Not this / boundary
overfitting	Adapting too closely to development examples so performance does not transfer well to new relevant cases.	
generalize	Perform usefully on new relevant cases beyond those used to develop a system.	
complexity	The amount of structure, work or resources a method requires, measured in a stated way.	
exception	A condition that requires handling outside the usual path.	

Words used in the closer explanation

Word	Meaning
Capacity	Flexibility of a model to represent patterns
Underfitting	Failing to capture useful structure
Generalization	Performance on relevant unseen examples

STUDENT LESSON · 6 OF 18

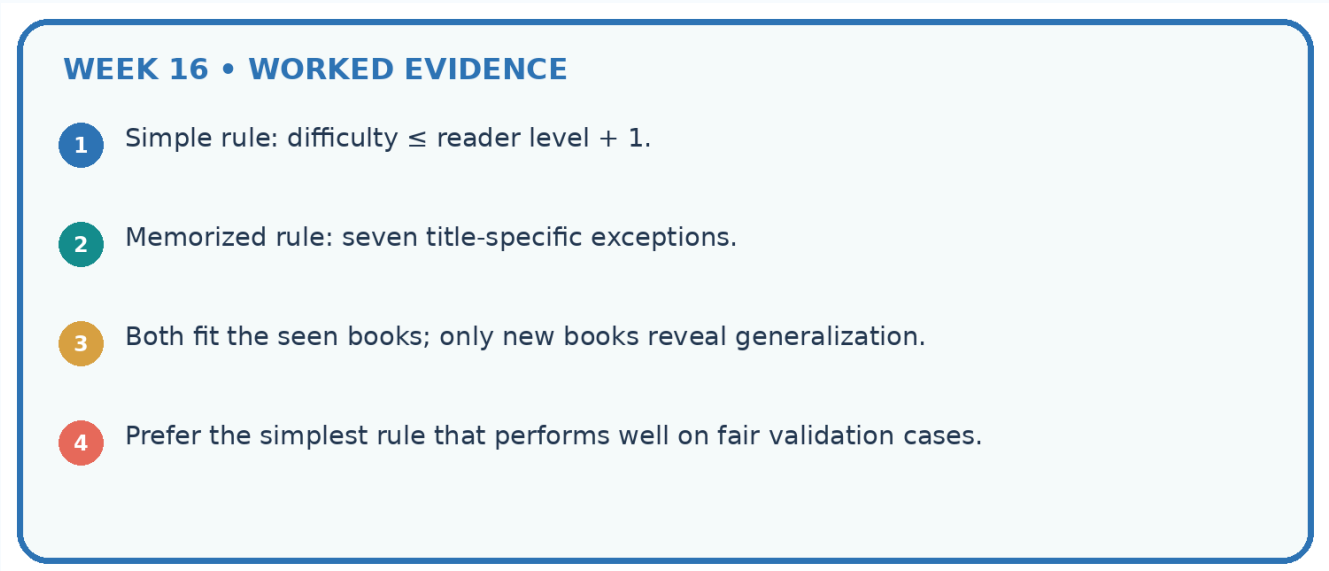
Follow a worked example

> Follow the worked trace for Overfitting Intuition, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.



1. Simple rule: difficulty ≤ reader level + 1.

2. Memorized rule: seven title-specific **exception**s.

3. Both fit the seen books; only new books reveal **generalization**.

4. Prefer the simplest rule that performs well on fair validation cases.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Overfitting and model capacity at a glance

PART	EVIDENCE OR ACTION	INTERPRETATION
Constant	Training 5	Validation 6
Line	Training 2	Validation 3
Flexible curve	Training 0	Validation 8

The curve result is consistent with overfitting, but a gap alone does not prove its cause. A changed population, bad labels or a split error could also produce a gap. Inspect the procedure before claiming a diagnosis. The constant may miss useful structure, but these numbers alone do not identify the true relationship.

Does a training-validation gap alone prove overfitting? Explain the steps or evidence behind your response.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Overfitting Intuition

A fictional project compares three book-length **prediction** models on the same development sets.

The facts we will use

Supplied facts

Mean absolute errors in pages: Simple **training** 5, validation 6; Medium **training** 2, validation 3; Complex **training** 0, validation 7.

Supplied facts

Select lowest validation error; untouched final tests are not used for selection.

AI ACADEMY / AI-2/16

Training and validation errors

	Training MAE pages	Validation MAE pages
Simple	5	6
Medium	2	3
Complex	0	7

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-2/16

Overfitting Intuition

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Training comparison

Complex fits training perfectly with zero error.



2. Validation comparison

Medium has error 3, versus Simple 6 and Complex 7.



3. Choose for development

Select Medium under the declared lowest-validation-error rule.



4. Final evaluation

Freeze the selection, then evaluate new reserved cases.

CHANGE ONE THING

Complex's validation error becomes 2 on a newly documented, fair development comparison.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Training comparison	Complex fits training perfectly with zero error.
2. Validation comparison	Medium has error 3, versus Simple 6 and Complex 7.
3. Choose for development	Select Medium under the declared lowest-validation-error rule.
4. Final evaluation	Freeze the selection, then evaluate new reserved cases.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

A flexible model can fit quirks of its **training** set without predicting new examples well. Here Complex's **training** success occurs alongside the worst validation error.

The numbers are invented comparisons, not a claim that **complexity** always hurts. Choosing Medium uses development **evidence**; its validation score must not be advertised as an untouched final-test result.

What this example does not establish

Keep the stated limit

A single small validation set may be noisy; selection **evidence** does not prove future population performance.

Keep the stated limit

Read the labelled picture: Does zero **training** error justify choosing Complex in the original record?

Keep the stated limit

Change one thing: Complex's validation error becomes 2 on a newly documented, fair development comparison. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Overfitting Intuition claim and explain a repair using the supplied evidence.

Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: A rule that is perfect on training data must be the best rule. Strong move: locate evidence, change one cause, and retest.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

A simpler model is not always better. If a flexible candidate has better valid **evidence** under the declared rule, keep that **evidence** visible.

Model **capacity** is one part of an experiment, alongside features, sample size, label quality and the split.

> **What extra evidence would you inspect before concluding a model memorized noise?**

STUDENT LESSON · 8 OF 18

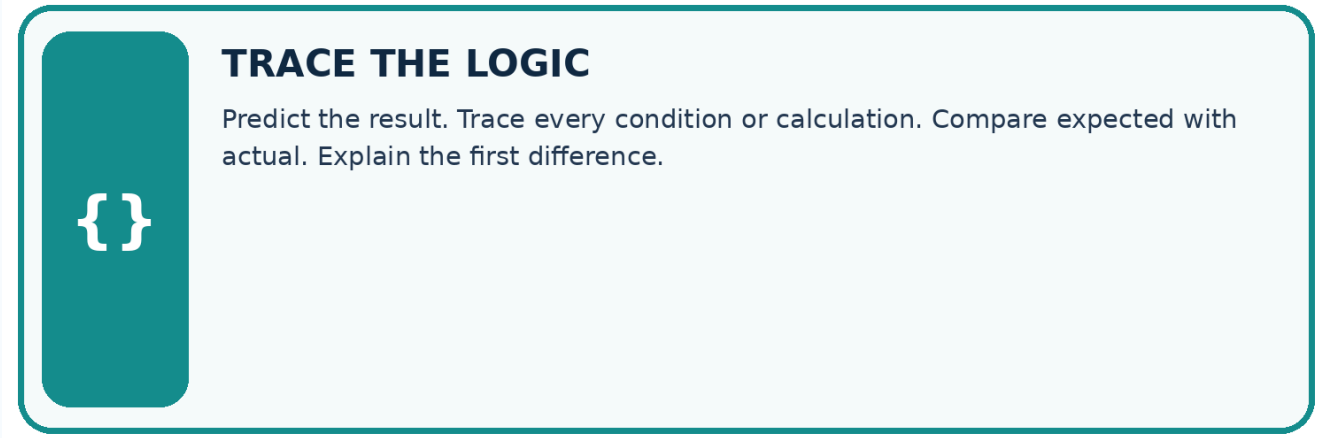
Practise together

> Practise Overfitting Intuition with support, then explain the deciding step in your own words.

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.



SET **evidence**_record PREDICT expected_result RUN one normal case and one boundary case COMPARE actual_result with expected_result IF different: locate first cause; change one factor; RETEST EXPLAIN result, limitation, and next safe step

> Trace one **input**. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

Change one development factor at a time: simplify the candidate, improve data coverage or adjust a declared regularization setting. Refit on **training** and compare on validation.

Do not remove difficult valid examples or repeatedly search the test for a favorable result.

Practise with less help: Interpret a gap

Use workbook W2 for the connected example "**Overfitting** and model **capacity**". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for **Overfitting** and model **capacity**. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

> Does zero **training** error justify choosing Complex in the original record?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

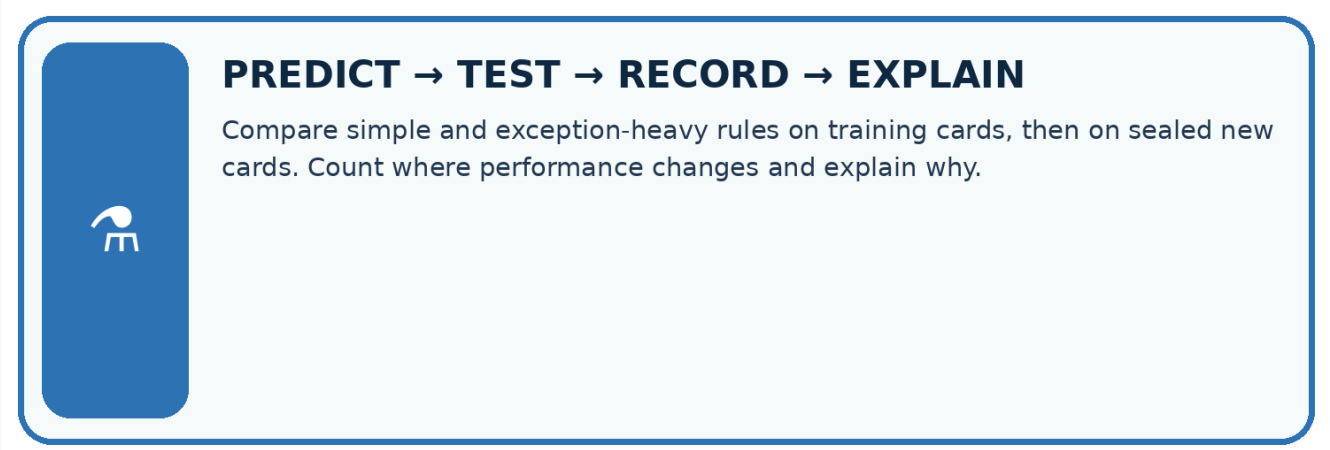
Run the investigation

> Investigate Overfitting Intuition. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Compare simple and **exception**-heavy rules on **training** cards, then on sealed new cards. Count where performance changes and explain why.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** boundary

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: missing

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

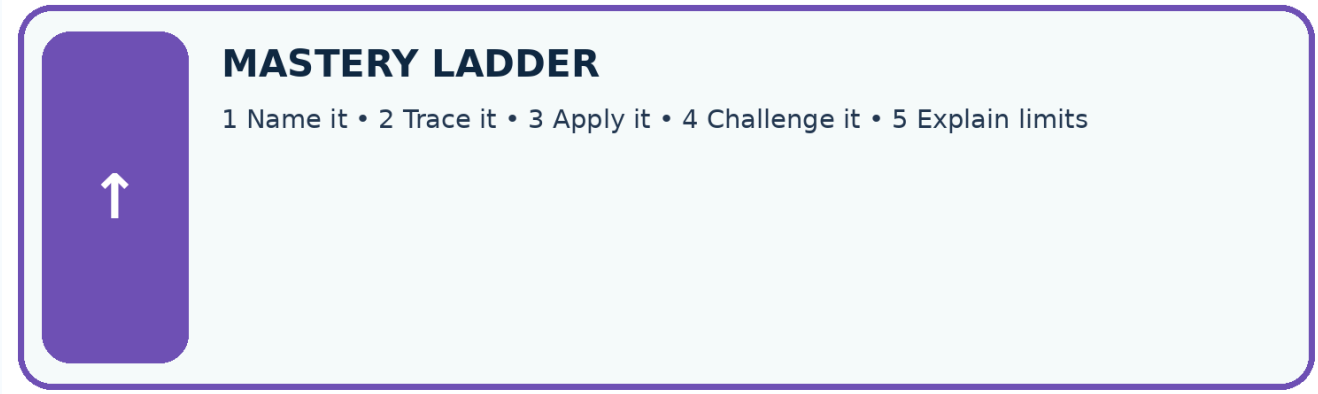
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Overfitting Intuition. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



1 • NAME: Define the concept without repeating the textbook.

2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.

3 • APPLY: Solve a different example independently.

4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest **evidence** of independent understanding:

> **Write or draw your reasoning and evidence.**

STUDENT LESSON · 11 OF 18

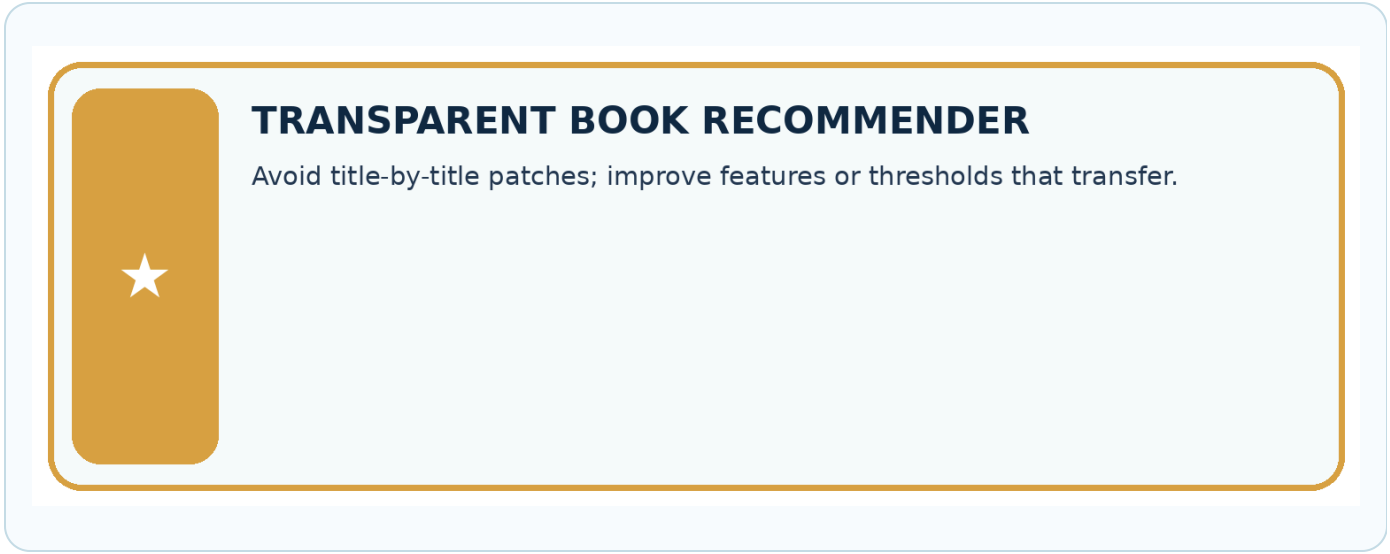
Connect your project

> Apply this week's idea to Transparent Book Recommender. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



THIS WEEK'S CHECKPOINT Avoid title-by-title patches; improve features or thresholds that transfer.

Trace the system

SYSTEM MAP Reader need → allowed **input**s → transparent rule or score → ranked candidates → explanation → test record → **human decision**

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under **human control**?

Write or draw your reasoning and evidence.

Responsible design

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse a ranking with a fact about a person.

Evidence for your project

Compare recommender revisions on fixed development cases, reserving new cases for the final presentation.

STUDENT LESSON · 12 OF 18

Show what you know

Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Independent evidence

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.

INDEPENDENT PROOF

Contrast a simple rule with a seven-exception rule.

Try a more demanding case

CHALLENGE Contrast a simple rule with a seven-**exception** rule.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My **evidence** that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

Check your explanation

SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Overfitting Intuition case for supported practice. Repeated examples do not become fresh assessments.

Week 16 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Rule A memorises six **training** book names. It scores 6/6 familiar and 2/6 new. Rule B uses relevant traits but is untested.

The question

> What does the **evidence** support?

Worked explanation

A shows weak transfer. B might generalise better but requires testing; simplicity is not proof.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Perfect recall can hide weak transfer

Evidence or stage	Value or result
Familiar books	6/6
New books	2/6

Rule A memorises six **training** book names. It scores 6/6 familiar and 2/6 new. Rule B uses relevant traits but is untested.

What does the **evidence** support?

A shows weak transfer. B might generalise better but requires testing; simplicity is not proof.

STUDENT LESSON · 14 OF 18

Try a new case independently

Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Apply independently

Models A **training** 4 validation 5, B **training** 1 validation 2, C **training** 0 validation 2. Lowest validation MAE then simpler; B is simpler than C. Choose and justify.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 16 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Rule A memorises six **training** book IDs. Rule B uses a relevant topic feature. Why might A fail on a new ID? What test is needed?

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Avoid title-by-title patches; improve features or thresholds that transfer.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

> Rule A memorises eight **training** IDs and gets 8/8 **training**, 2/8 validation. Rule B uses genre and gets 6/8 **training**, 6/8 validation. Which is preferable on this **evidence**, and what test remains?

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Overfitting Intuition, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: A fictional project compares three book-length **prediction** models on the same development sets.

By the end, I can	How I will show it
Compare training and validation evidence for model capacity	Explain the deciding step, show a trace or test, and state one limit.

By the end, I can	How I will show it
Explain overfitting without assuming every gap proves it	Explain the deciding step, show a trace or test, and state one limit.
Use a declared selection rule instead of perfect training fit	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Overfitting Intuition without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Compare **training** and validation **evidence** for model **capacity**

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional project compares three book-length **prediction** models on the same development sets.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Freeze the selection, then evaluate new reserved cases.

1. Final evaluation

2. **Training** comparison

3. Choose for development

4. Validation comparison

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Final evaluation Freeze the selection, then evaluate new reserved cases. A flexible model can fit quirks of its **training** set without predicting new examples well.

Here Complex's **training** success occurs alongside the worst validation error. The numbers are invented comparisons, not a claim that **complexity** always hurts.

Choosing Medium uses development **evidence**; its validation score must not be advertised as an untouched final-test result.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Complex's validation error becomes 2 on a newly documented, fair development comparison. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. This one example guarantees correct results for every future **input**.

3. The stated rule now selects Complex.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The stated rule now selects Complex. The rule compares validation errors; it does not ban complex models.

Repair a missing building block

Review Choose Informative Questions (ai-2/15). Goal: Fit a root question and leaf **prediction**s from **training** examples. Short practice: Choose the best original question and explain why its more balanced split matters here..

Return to this week and explain what you changed.

Review Types of Classification Error (ai-2/13). Goal: Build a labelled confusion matrix from matched reference and **prediction** pairs. Short practice: Which error causes a sound book to be sent for unnecessary repair review?.

Return to this week and explain what you changed.

Review Human Classifier Simulation (ai-2/10). Goal: Construct a classifier with a stated **input** format, selected rule and recorded **output**.

Short practice: Why are J and M initially treated differently even though both are illustrated mysteries?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

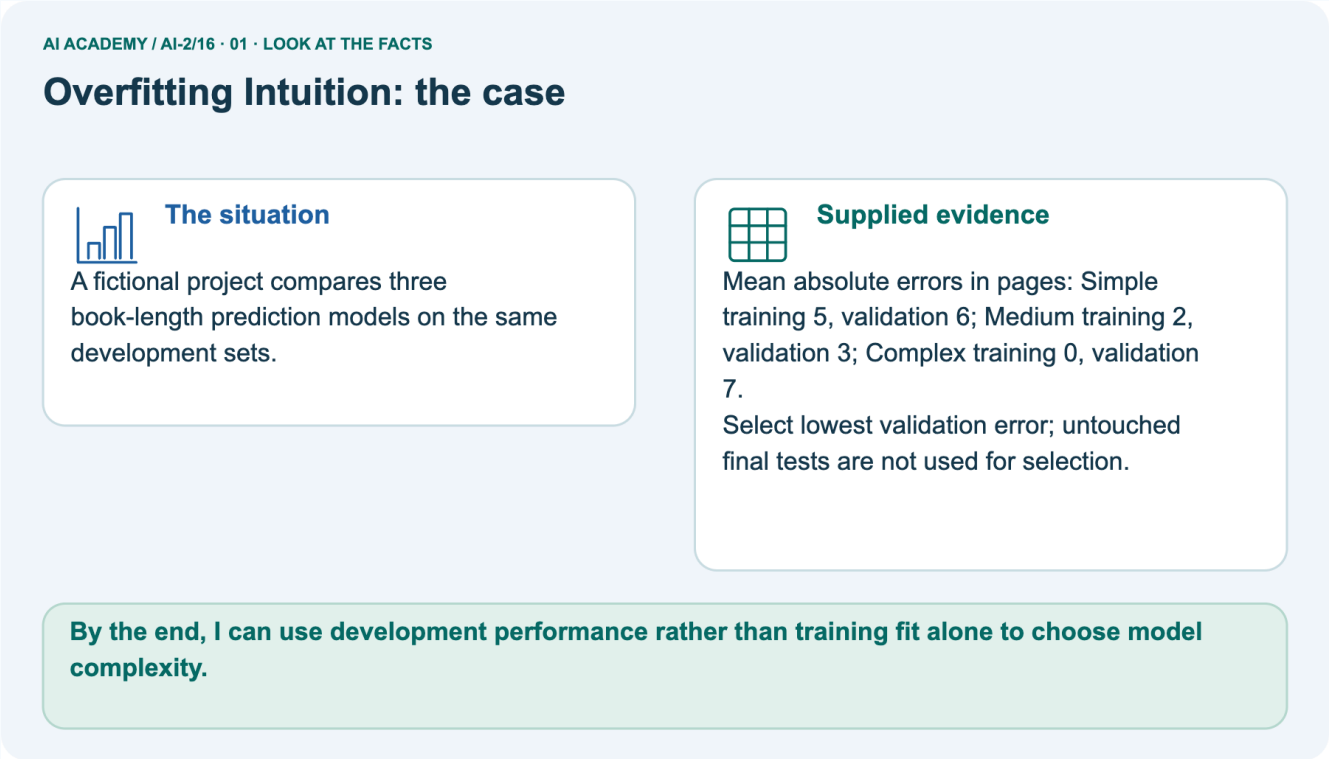
Drafts stay in this browser when storage is available; clear drafts on a shared device.

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Visual concept journey

> Explain each Overfitting Intuition graphic using its labels, deciding evidence and relevant limitation.

01. Overfitting Intuition: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

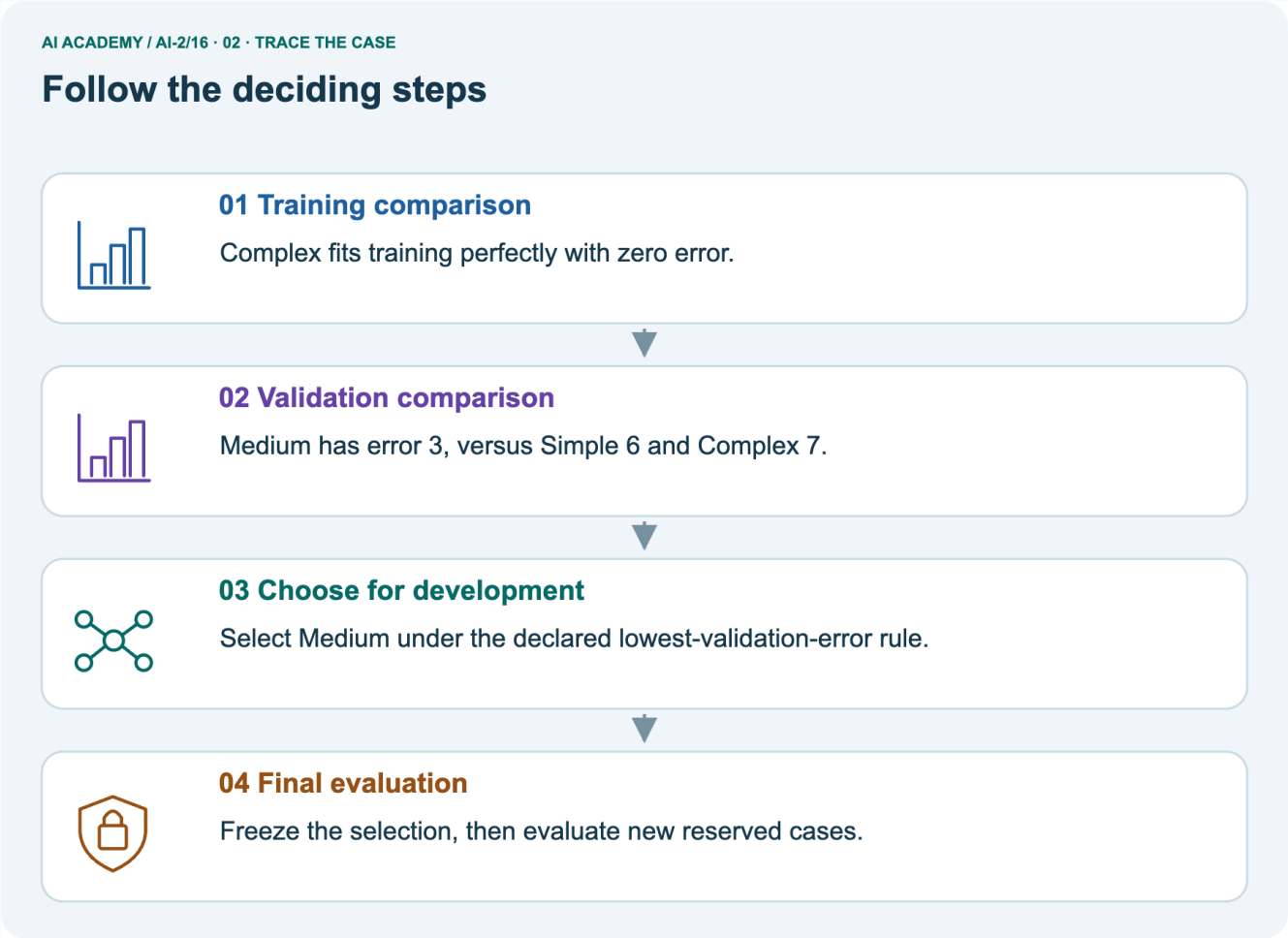
A fictional project compares three book-length **prediction** models on the same development sets.

Supplied facts: Mean absolute errors in pages: Simple **training** 5, validation 6; Medium **training** 2, validation 3; Complex **training** 0, validation 7.

Select lowest validation error; untouched final tests are not used for selection.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

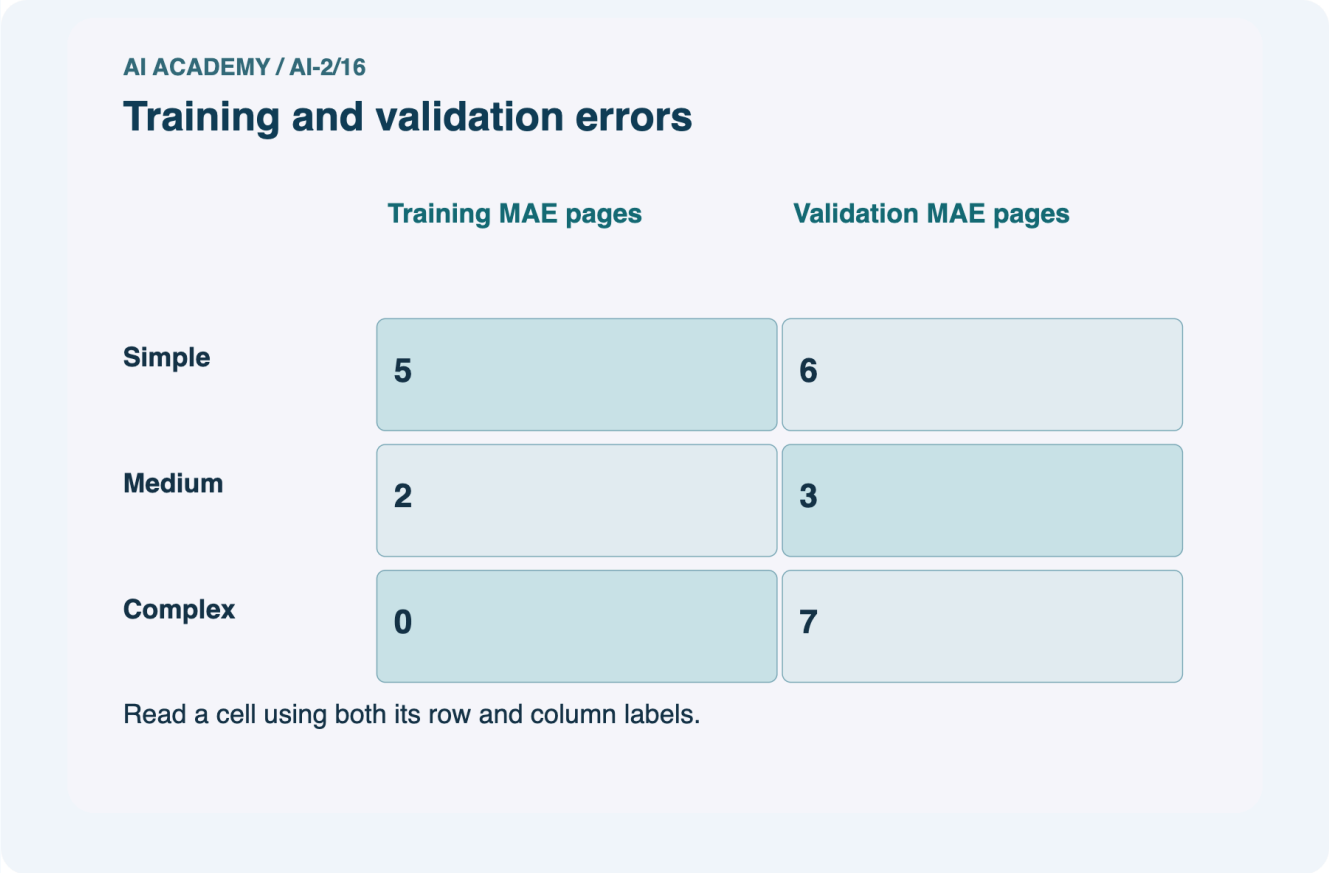


Follow the labelled picture with the matching explanation. Open the vector diagram

Training comparison: Complex fits **training** perfectly with zero error. → Validation comparison: Medium has error 3, versus Simple 6 and Complex 7. → Choose for development: Select Medium under the declared lowest-validation-error rule. → Final evaluation: Freeze the selection, then evaluate new reserved cases.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Training and validation errors

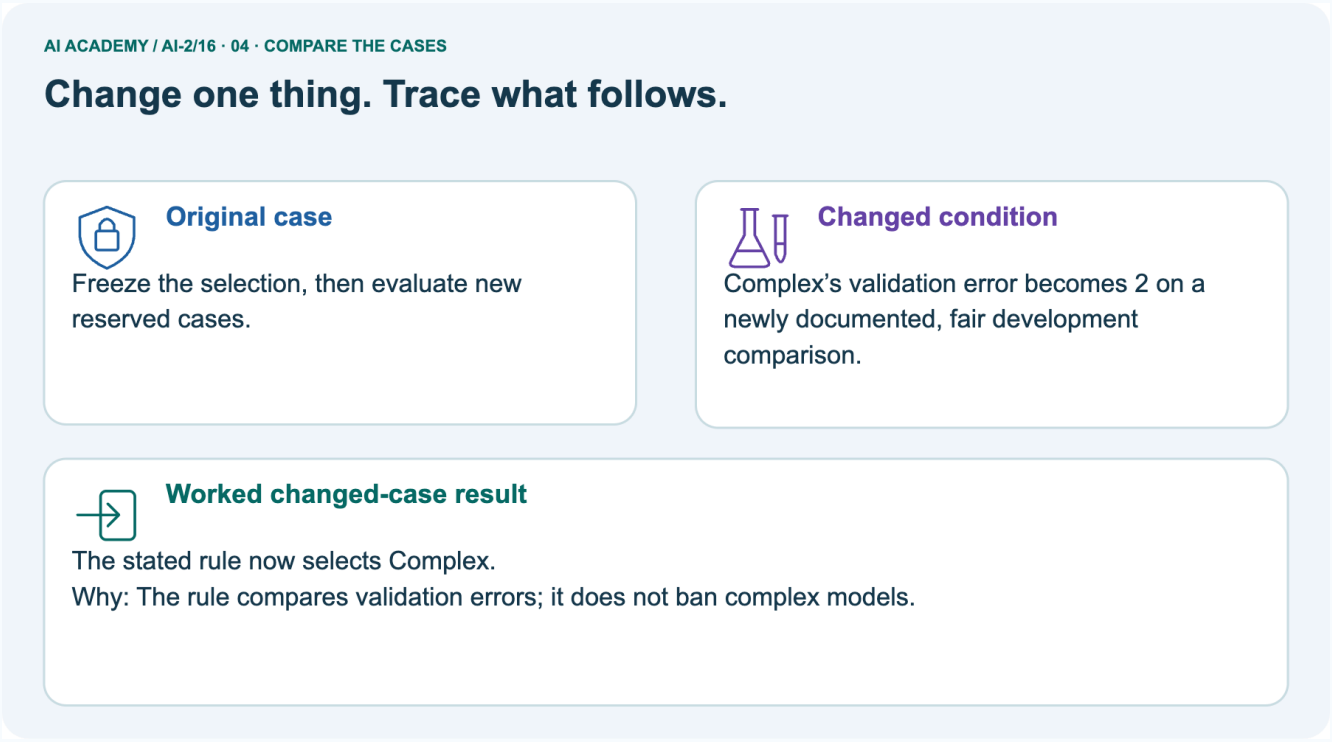


Follow the labelled picture with the matching explanation. Open the vector diagram

Training and validation errors. Simple: **Training** MAE pages = 5, Validation MAE pages = 6. Medium: **Training** MAE pages = 2, Validation MAE pages = 3. Complex: **Training** MAE pages = 0, Validation MAE pages = 7.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



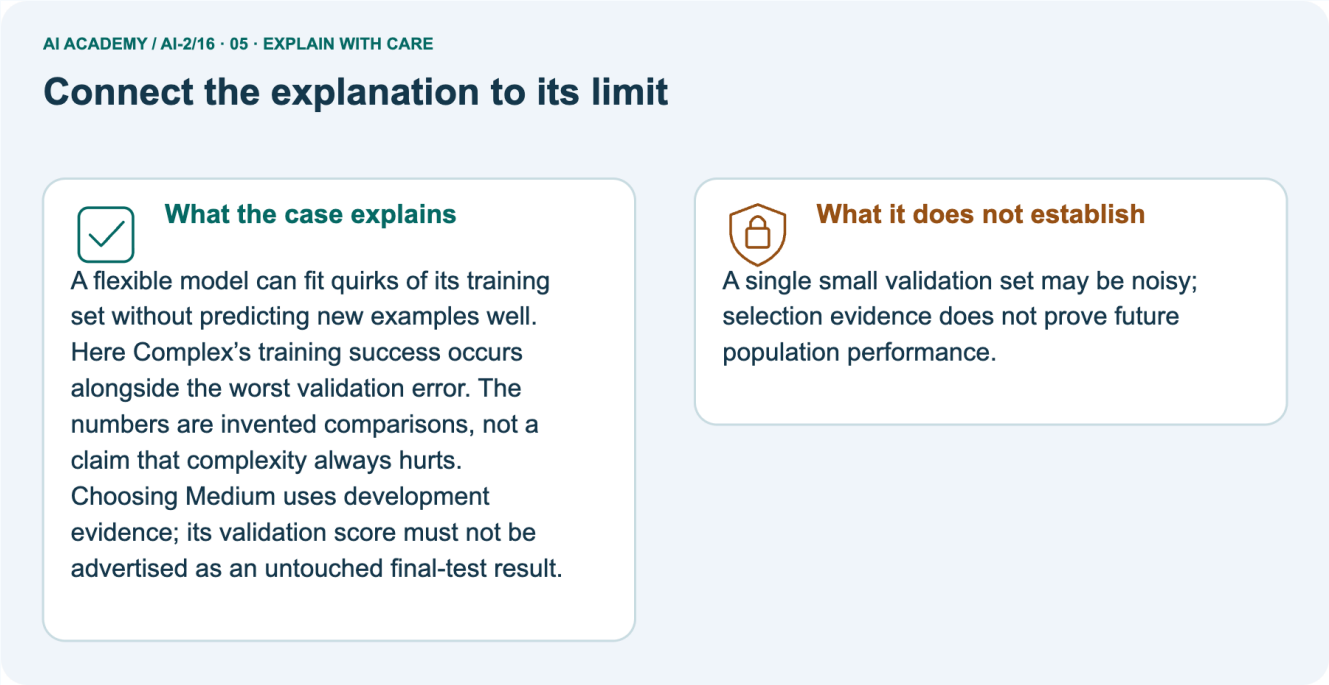
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Freeze the selection, then evaluate new reserved cases. Changed condition: Complex's validation error becomes 2 on a newly documented, fair development comparison.

Expected result: The stated rule now selects Complex. Reason: The rule compares validation errors; it does not ban complex models.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

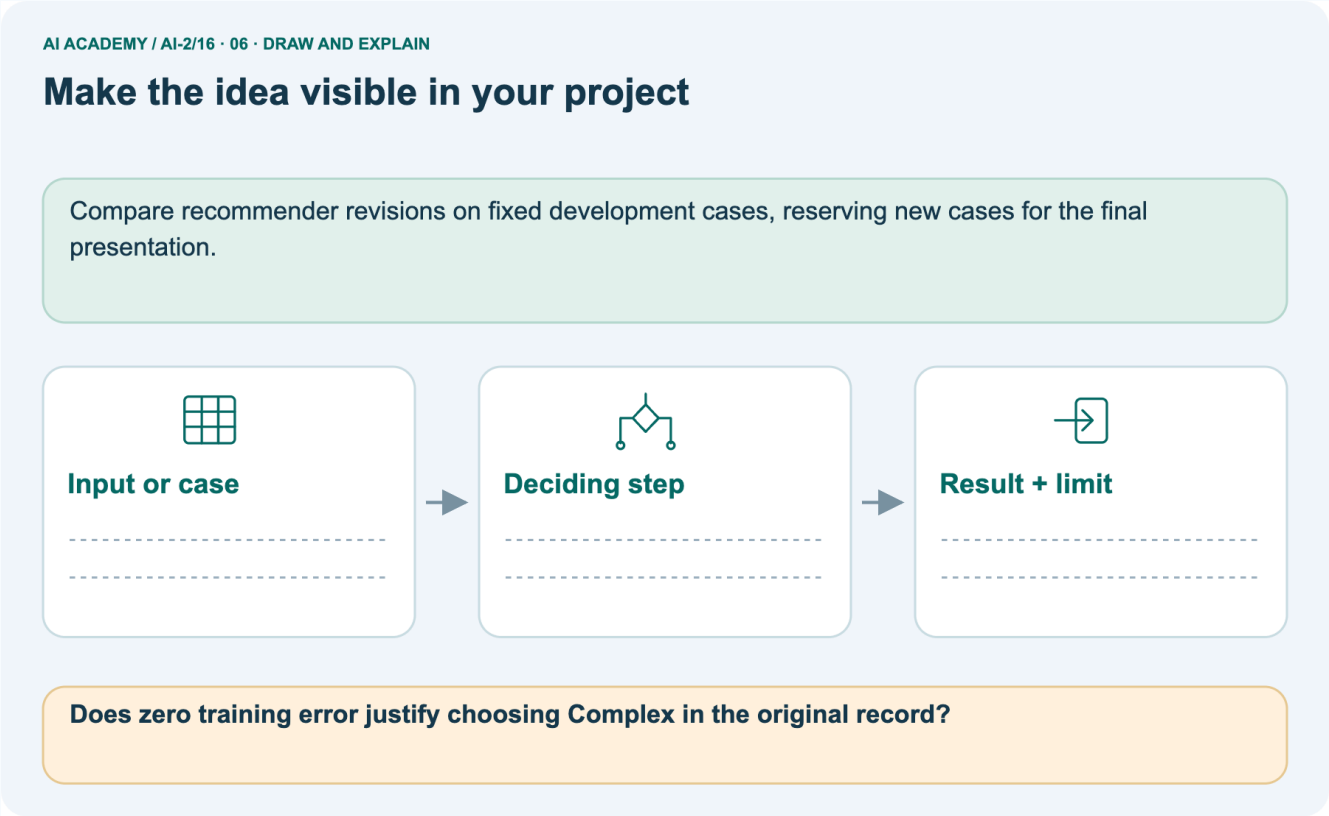
A flexible model can fit quirks of its **training** set without predicting new examples well. Here Complex's **training** success occurs alongside the worst validation error.

The numbers are invented comparisons, not a claim that **complexity** always hurts. Choosing Medium uses development **evidence**; its validation score must not be advertised as an untouched final-test result.

Limit: A single small validation set may be noisy; selection **evidence** does not prove future population performance.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Compare recommender revisions on fixed development cases, reserving new cases for the final presentation. Draw your **input**, deciding step, result and limit.

Supported question: Does zero **training** error justify choosing Complex in the original record?

Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.